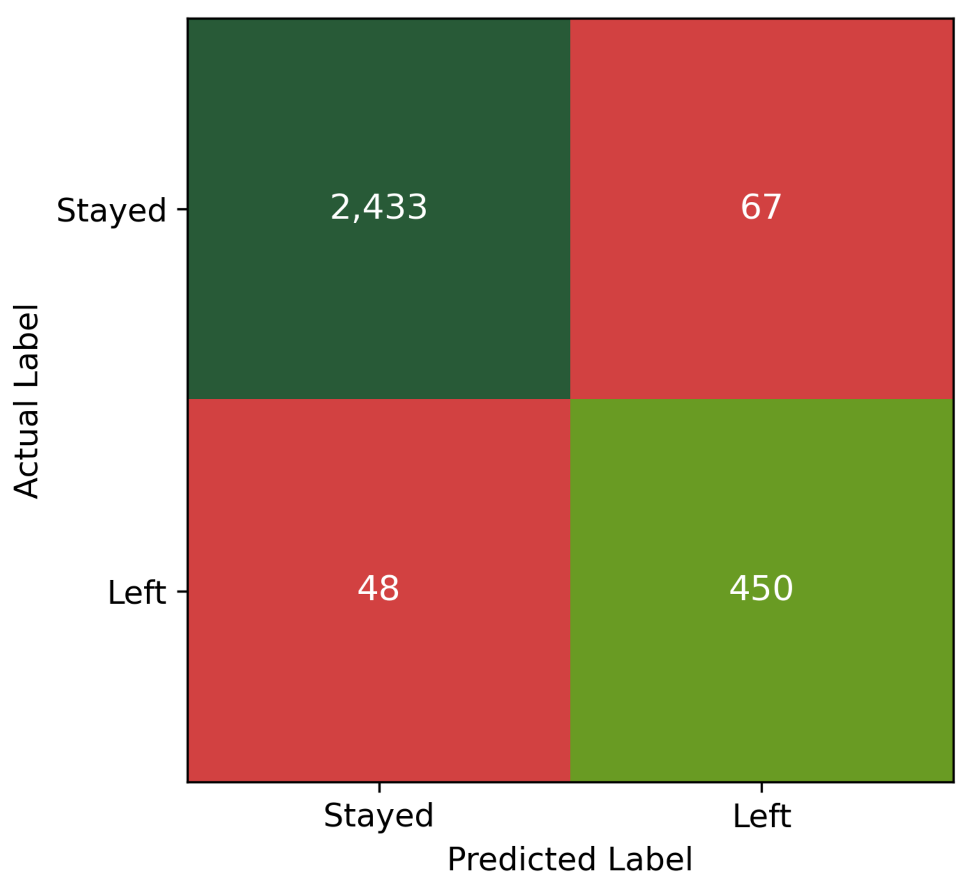
Salifort Motors: Employee Attrition Analytics

Identifying Employees at Risk of Leaving Using Machine Learning

Model Performance Comparison

Model Name	Recall	Precision	F1 Score	Accuracy
Random Forest (Champion)	90.36%	87.04%	88.67%	96.16%
Decision Tree	91.77%	78.39%	84.55%	94.43%
Logistic Regression	26.00%	44.00%	33.00%	82.00%

Random Forest Confusion Matrix



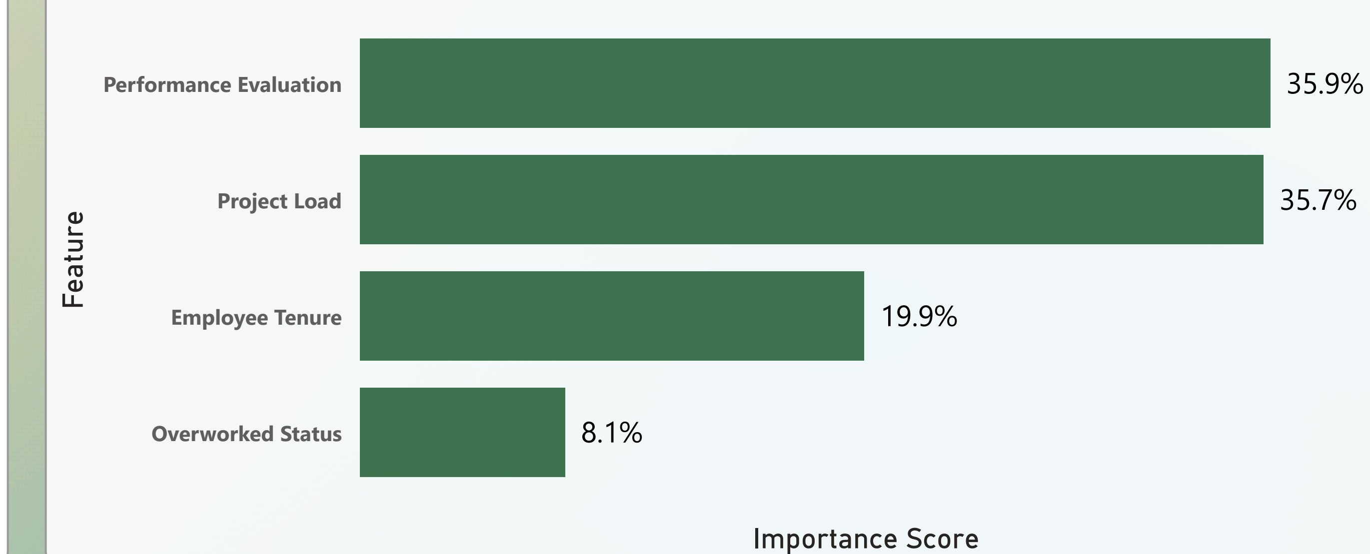
Business Impact of the Model

- **High Detection Rate:** Successfully identified **450 (90%) of employees who ultimately left**, missing only **48 attrition cases**.
- **Strategic Trade-off:** Prioritizes catching costly flight risks while keeping false alarms (**67 false positives**) at a highly manageable level.
- **Resource Optimization:** Correctly identified **2,433 stable employees**, ensuring that retention budgets and HR efforts are focused precisely where they are needed.
- **HR Enablement:** Empowers leadership to shift away from reactive backfilling toward proactive retention strategies.

Model Selection Insight

Random Forest was selected as the **champion model** due to its strong balance of **Recall (90.36%)**, **Precision (87.04%)**, and **F1 Score (88.67%)**. While the Decision Tree achieved slightly higher Recall, Random Forest delivered better overall predictive performance and a **lower false-positive rate**, making it the preferred model for employee retention strategies.

Feature Importance Drivers



Analytical Conclusion & Readiness

The machine learning phase validates that employee turnover at Salifort Motors is a **predictable operational challenge** rather than an unpredictable human factor. With **96.16% Accuracy**, **90.36% Recall**, and **87.04% Precision**, the **Champion Random Forest model** serves as a reliable early-warning system for attrition. By identifying **performance evaluation**, **project load**, **employee tenure**, and **overworked status** as the primary drivers of turnover, the analysis provides a foundation for **targeted retention strategies**.

Feature Importance Insight

Performance Evaluation (35.9%) and **Project Load (35.7%)** emerged as the strongest attrition drivers, followed by **Employee Tenure (19.9%)** and **Overworked Status (8.1%)**. These findings indicate that **workload intensity**, **performance pressure**, and **tenure-related factors** have a greater influence on employee turnover than compensation or department-level differences.